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## Education

**University of Illinois** Urbana-Champaign, College of Liberal Arts and Sciences Aug 2023 - Present  
*Major in Statistics & Computer Science — Cumulative GPA: 3.98 / 4.00*

**Henry M. Gunn High School** Palo Alto, California Aug 2019 - May 2023  
*Cumulative GPA: 4.33 / 4.00*

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## Publications

- Vu-Le, T., Park, M., **Chen, I.**, Chacko, G., Warnow, T. (2025). Using Stochastic Block Models for Community Detection: The issue of edge-connectivity. *Applied Network Science*. 10.1007/s41109-025-00747-2
  - Vu-Le, T., Lamy, J., **Chen, I.**, Park, M., Harb, E., Chacko, G., Warnow, T. (2025). Dense Subgraph Clustering and a New Cluster Ensemble Method. *Complex Networks* 2025. 10.48550/arXiv.2508.17013v2
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## Technical Skills

### Programming Languages; Tools

Python, C++, C#, R, Rust, Bash, SQL, MIPS; Git, L<sup>A</sup>T<sub>E</sub>X

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## Work Experience

### Siebel School of Computing and Data Science

*Undergraduate Research Assistant* May 2025 - Present

- Conducted a literature review comparing state-of-the-art (SotA) community detection (CD) algorithms
- Designed experiments demonstrating that a novel CD algorithm outperforms the SotA alternatives.
- Implemented a new highly parallizable community search algorithm on a high performance system with terabit memory and 128-cores computing nodes.
- Edited the manuscripts for two publications accepted in peer reviewed conferences and journals.

### Maven Optronics

May 2024 - Aug 2024

*Software Engineer Intern*

- Implemented Manufacturing Execution System (MES) materials traceability modules — populated mySQL tables to link raw material barcode IDs (with photo records) to assembled MiniLED boards.
  - Developed automatic deployment of software releases via win32 installer (.msi) through administrative installation across NAS server.
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## Projects

**ClassFS: course explorer filesystem** (<https://github.com/Yasmercy/ClassFS>) Aug 2024 - Dec 2024

- Collaborated with two team members to build virtual file system through FUSE, allowing users to browse the UIUC course catalog with their file explorer of choice.
- Implemented system calls hooks for `open`, `read`, `readdir`, and `stat`.

**Directle: NLP-enabled linguistics game** (<https://github.com/Yasmercy/Directle>) Jan 2024 - May 2024

- Collaborated with three team members to build a full-stack website hosting an English word game. The game guides players to guess a secret word through one-word semantic clues.
  - Implemented a word2vec NLP model using the top 10M wikipedia articles to calculate of word similarities.
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## Relevant Coursework

**Computer Science:** *Communication Networks; Operating Systems; Computer Architecture I & II; Algorithms I & II; Randomized Algorithms; Computational Biology*

**Statistics & Math:** *Probability I & II; Modeling I & II; Numerical Analysis; Graph Theory; Linear Algebra; Differential Equations; Real Analysis; Abstract Algebra*

**Physics:** *Mechanics; E&M; Thermodynamics; Quantum Physics; Special Relativity*